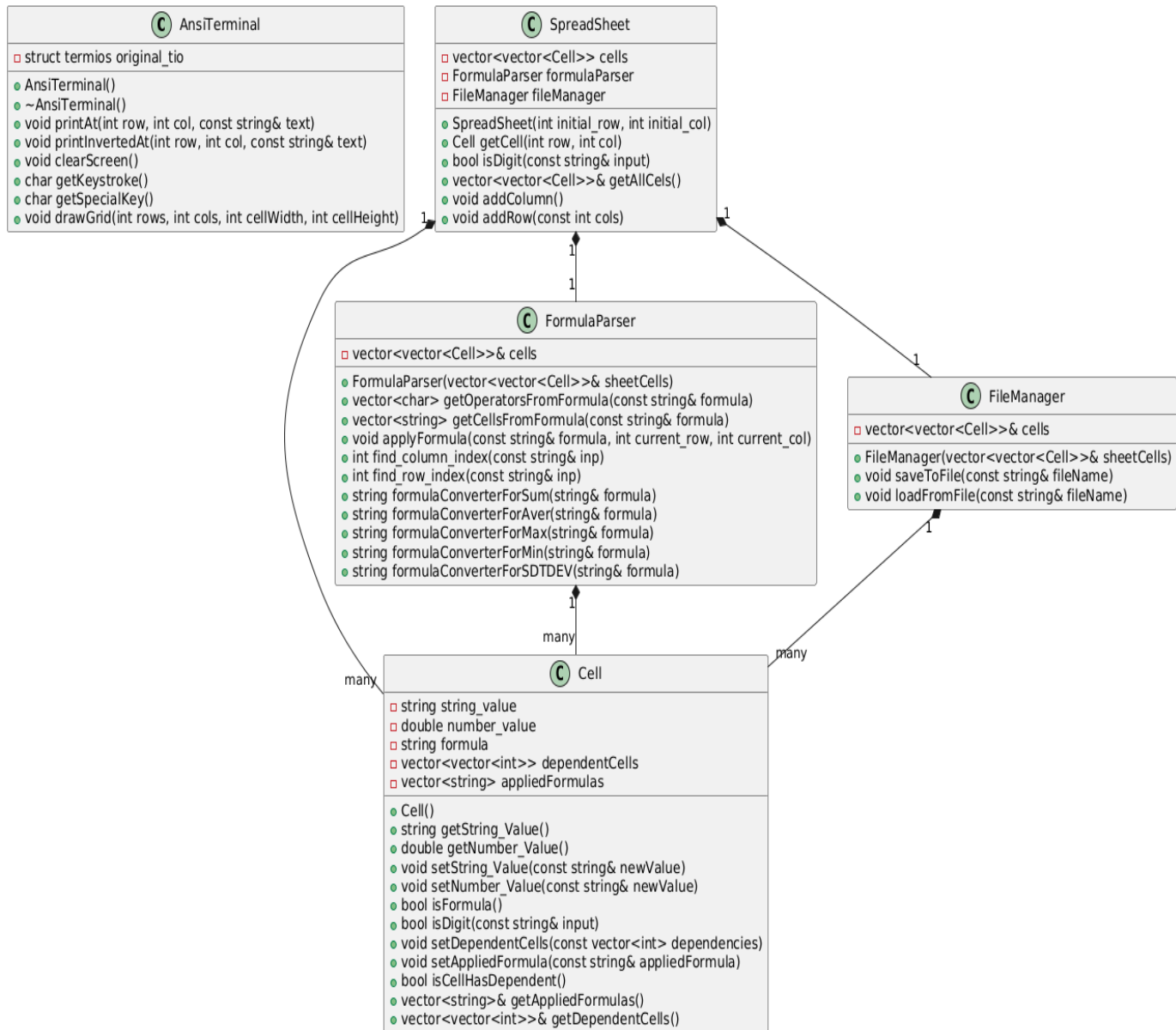


UML DIAGRAM:



SUM FUNCTION:

[illegible][illegible]

MAX FUNCTION:

[illegible]

MIN FUNCTION:

Active Cell: Cell [C4]: =MIN(A1..A4)

[illegible]

Active Cell: Cell [C4]:

SDTDEV FUNCTION:

Active Cell: Cell [D3]: =STDDEV(A1..A4)

[illegible]

Active Cell: Cell [D3]:

AVER FUNCTION:

[illegible]

SOME OPERATIONS:

1-

[illegible]

[illegible]

[illegible][illegible]

4-

[illegible][illegible]

5-

[illegible][illegible]

Implemented Features:

Core Spreadsheet Management

1. Cell Management:

- Stores string values (string_value).
- Stores numeric values (number_value).
- Supports formulas stored as strings (formula).
- Manages dependent cells (dependentCells) and applied formulas (appliedFormulas).

2. Row and Column Operations:

- addRow: Adds a new row with the specified number of columns.
- addColumn: Adds a new column to all existing rows.

3. Grid Access and Management:

- Retrieves all cells or a specific cell (getAllCells, getCell).

Implemented Features

Below is a list of the implemented features based on the provided files and descriptions:

Core Spreadsheet Management

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- Stores string values (string_value).
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- addRow: Adds a new row with the specified number of columns.
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- Retrieves all cells or a specific cell (getAllCells, getCell).
-

Formula Handling

1. Formula Parsing:

- Extracts operators (+, -, *, /) from a formula.
- Identifies cell references (e.g., A1, B2).

2. Formula Application:

- Applies formulas to a specific cell (applyFormula).
- Handles dependencies between cells.

3. Specialized Formula Support:

- **SUM**: Converts and applies sum formulas (e.g., SUM(A1..A3) to A1+A2+A3).
- **AVER**: Calculates averages of ranges (e.g., AVER(A1..A3)).
- **MAX**: Finds the maximum value in a range (e.g., MAX(A1..A3)).
- **MIN**: Finds the minimum value in a range (e.g., MIN(A1..A3)).
- **STDDEV**: Calculates the standard deviation for a range (e.g., STDDEV(A1..A3)).

File Operations

1. Saving:

- Saves the current spreadsheet state to a CSV file.

2. Loading:

- Loads spreadsheet data from a CSV file, populating the grid.

Terminal-Based User Interface

1. Grid Visualization:

- Renders a grid in the terminal with adjustable dimensions (drawGrid).

2. Cell Interaction:

- Displays and edits cell values.
- Supports keyboard navigation (arrow keys for movement).
- Handles user inputs for formulas and values.

3. Keyboard Shortcuts:

- l: Save the spreadsheet to a file.
- q: Quit the program.
- J: Apply formulas or save input to cells.

4. Dynamic Grid Resizing:

- Adds new rows or columns dynamically when navigating beyond the current boundaries.

USER MANUAL:

How to Run:

1-Compile the program:

```
g++ -o spreadsheet main.cpp SpreadSheet.cpp Cell.cpp FileManager.cpp  
FormulaParser.cpp AnsiTerminal.cpp
```

2- Run the program:

```
./spreadsheet
```

Keyboard Shortcuts:

Key	Action
Arrow Keys	Navigate between cells (Up, Down, Left, Right).
I	Save the current spreadsheet to a file (<code>output.csv</code>).
q	Quit the program.
J (Enter)	Save entered data or apply a formula to the selected cell.
Backspace	Remove the last character of the input.
P	Load the spreadsheet from a file (<code>test_spreadsheet.csv</code>).

How to Use:

1. Navigating the Grid

- Use the **arrow keys** to move between cells:
 - **Up (U)**: Moves to the cell above.
 - **Down (D)**: Moves to the cell below.
 - **Left (L)**: Moves to the cell on the left.
 - **Right (R)**: Moves to the cell on the right.
- If you navigate beyond the current boundaries:
 - **New Row**: Automatically added when moving beyond the last row.
 - **New Column**: Automatically added when moving beyond the last column.

2. Editing Cells

- **Entering Values:**
 1. Type a value directly into the current cell.
 2. Press **J (Enter)** to save the value in the cell.
- **Deleting Values:**
 - Move to the desired cell.
 - Press **Backspace** until the value is cleared.
 - Press **J** to confirm the deletion.

3. Using Formulas

Formulas start with an equal sign (=). Supported formulas include:

Formula	Example	Description
SUM	<code>=SUM(A1..A3)</code>	Calculates the sum of cells A1 to A3.
AVER	<code>=AVER(B1..B3)</code>	Calculates the average of cells B1 to B3.
MAX	<code>=MAX(C1..C3)</code>	Finds the maximum value in cells C1 to C3.
MIN	<code>=MIN(D1..D3)</code>	Finds the minimum value in cells D1 to D3.
STDDEV	<code>=STDDEV(E1..E3)</code>	Calculates the standard deviation of cells E1 to E3.

4. Saving Data

- Press **I** to save the current spreadsheet to output.csv.
- The program overwrites the file with the latest data.

5. Loading Data

- When the program starts, it automatically loads data from output.csv if the file exists.

AI-Based Assistance:

- 1- I used REGEX with the help of ChatGPT.
- 2- I sought help from ChatGPT to improve code readability and make some enhancements.
- 3- I used ChatGPT's assistance to create the UML diagram.
- 4- I received help from ChatGPT while preparing this PDF.
- 5- I utilized ChatGPT's guidance to draw the grid in the terminal.

The program's shortcomings:

- 1- When adding rows or columns that exceed the screen's capacity, the table layout breaks, causing display issues.
- 2- Some improvements can be made, such as enhancing error handling to make the program more robust.
- 3- There is an issue with the table breaking while reading data from a file named "test_spreadsheet.csv". I couldn't figure out the reason for this.